

Week 2 Control a character

Challenge • GROW computing • 40 minutes

Name: _____ Date: _____

Make right and left controls. Test, save and reopen.

Point, say, draw or write. Someone may record your exact words.

Arrival → Starter → I Do → We Do

I Do 2 → We Do 2 · Task → Independent Task → Exit

Arrival 0–3 min

Open Scratch.

Load W02_Start.sb3.

Starter 3–6 min

The left key has: change x by 10.

Will it go left or right? Left / Right

I Do 6–11 min

Watch the teacher. Point to the block that starts when a key is pressed.

We Do 11–18 min

1. Make your prediction.
2. Ask a partner or adult to check your idea.
3. Test it together.
4. Point to what explains the result.

Help to fix the left script in our shared model.

Reset the model to the centre.

Press once	I expect x	I see x
Left		
Then right		

Our reason or the block we checked

My core task and challenge

Choose one investigation once your core task works

I Do 2 18–21 min

Watch the y control model.

We Do 2 • Task 21–25 min

Predict and explain a route to (20,20). Test the shared idea.

Independent Task 25–35 min

First, create right and left scripts in W02_Start.sb3.
Reset. Test right, right, left, left. Explain the result.

Investigation A Two routes

Add up and down scripts. Reset to (0,0).
Reach (20,20). Reset before trying another order.

Route	My key order	Finish
First		
Second		

Why do both work in this open space?

Investigation B Smaller moves

Change horizontal moves to +5 and -5. Reset.
Predict: how many right presses reach $x = 20$?

My prediction	Actual presses

Reset to (0,0). Test right, right, left, left.
Explain its finish. Restore moves to +10 and -10.

Save and explain your thinking

You may explain aloud instead of writing

Exit 35–40 min

- ☐ Choose File.
- ☐ Choose Save to your computer.
- ☐ Use the name W02_MyControls.sb3.
- ☐ Find it in your school computing folder.
- ☐ Choose File, then Load from your computer.
- ☐ Open the file you just saved.
- ☐ Run it. Check that your change is still there.

If it downloads to Downloads, ask for help to put it in your school folder.

I tested this idea

The result happened because

What I said and what it changed

My mission: Space Station / Delivery Depot / Wildlife Ranger

A feature I would like

Teacher response: use it / change it with me / try it later

What changed or the reason for waiting

You can say this to an adult. They can write your words.

Useful reference

Right: change x by 10. Left: change x by -10.

Up: change y by 10. Down: change y by -10.

Press and release once for each test input.